

From separate-beam readout to one-beam metrology with Raman loss

Abstract

We describe the progression of the model used to estimate a three-component field from a randomly controlled spin ensemble: separate-beam readout with spin flips but no loss (v5), one-beam readout with spin flips (v6), and the current one-beam model with cesium-derived coefficients and Raman loss (v7). We specify the implemented equations, cluster configuration, and analysis of the four v7 datasets. The central comparison switches the tensor Hamiltonian on and off at fixed scattering and readout noise. The saved results show shorter times to a target filtered variance with the tensor term, with gains that depend on the target, probe strength, and optical depth.

1 Common estimation problem and model progression

We model one spin J in an initially uncorrelated ensemble of N_a atoms. The unknown static field is $\mathbf{w} = (w_x, w_y, w_z)$. With $\hbar = 1$,

$$H(t) = \mathbf{w} \cdot \mathbf{J} + \Omega[\cos \phi(t)J_x + \sin \phi(t)J_y] + \frac{\beta_H}{2J}J_z^2. \quad (1)$$

The factor $1/(2J)$ is part of the code convention: `beta` is not the coefficient multiplying J_z^2 directly. Frequencies are in units of Ω , time is plotted as Ωt , and the scripts set $\Omega = 1$. The phase is uniform on $[0, 2\pi)$, constant for blocks of 1,000 steps, and independently drawn for each realization. The runs discussed here use $J = 3$, $\rho_0 = |J, +x\rangle\langle J, +x|$, and observable $O_0 = J_y$.

A record contains a known control sequence and a signal normalized to the *initial* atom number:

$$M_n = s_n(\mathbf{w}) + \xi_n, \quad \xi_n \sim \mathcal{N}(0, \sigma^2), \quad s_n = \text{Tr}[O_n(\mathbf{w})\rho_0]. \quad (2)$$

Noise samples are independent within a record. The forward model evolves an ensemble expectation, rather than individual scattering trajectories or a conditional collective quantum state.

Model	Readout	Irreversible dynamics	Probe sweep
v5	Separate beam; fixed σ	Spin flips; no manifold loss	$\gamma = \beta/\eta$
v6	Same beam as tensor shift	Spin flips; no manifold loss	γ and σ tied to β
v7	Same beam as tensor shift	Spin flips and Raman loss	Atomic coefficients set rates and σ

Table 1: The model progression. The closed-system v4 script is the zero-dissipation predecessor. “No loss” in v5/v6 means trace preservation, not absence of decoherence.

2 Separate beams without manifold loss: v5

2.1 Scattering from the tensor beam

The first open-system model retains two within-manifold Raman spin-flip operators,

$$L_+ = \sqrt{\gamma}J_+, \quad L_- = \sqrt{\gamma}J_-, \quad J_{\pm} = J_x \pm iJ_y, \quad \gamma = \frac{\beta}{\eta}, \quad \beta_H = \beta. \quad (3)$$

The master equation is

$$\dot{\rho} = -i[H(t), \rho] + \mathcal{D}_{\text{sf}}[\rho], \quad (4)$$

$$\mathcal{D}_{\text{sf}}[\rho] = \gamma \sum_{\nu=\pm} \left(J_{\nu} \rho J_{\nu}^{\dagger} - \frac{1}{2} \{ J_{\nu}^{\dagger} J_{\nu}, \rho \} \right) = -\gamma \sum_{k=x,y} [J_k, [J_k, \rho]]. \quad (5)$$

This channel preserves trace and is unital. Its adjoint has the same operator form, with exact dissipative identities

$$\mathcal{D}_{\text{sf}}^{\dagger}[J_x] = -\gamma J_x, \quad \mathcal{D}_{\text{sf}}^{\dagger}[J_y] = -\gamma J_y, \quad \mathcal{D}_{\text{sf}}^{\dagger}[J_z] = -2\gamma J_z. \quad (6)$$

Thus γ is a transverse spin-relaxation coefficient, not the total photon scattering rate. The tensor Hamiltonian couples first moments to higher moments; these identities do not close the full driven dynamics.

Far from the excited hyperfine structure, the tensor shift and scattering both scale as intensity divided by detuning squared. This motivates a fixed ratio $\eta = \beta/\gamma$; the original run chose $\eta = 30$. This phenomenological spin-flip ratio differs from the total-scattering ratio η_s introduced below. The optical-pumping and light-shift framework is described in Ref. [1]; our numerical conventions are specified by the repository scripts.

2.2 Independent readout and its limitation

A separate readout beam supplies N_p photons per interval and Faraday gain G per atom per unit spin. The model uses

$$\sigma^2 = \frac{1}{N_p G^2 N_a^2}, \quad (7)$$

independently of β , and omits readout-induced scattering. The dataset `data_mle_over_time_decoherence_1` uses $\sigma = 0.021$ and $\beta \in \{0, 0.5, 1, 2, 5\}$. At $\beta = 0$, both the tensor shift and modeled spin flips vanish, while finite-noise readout remains available. The reference is therefore an idealized decoherence-free measurement.

A physical readout beam also scatters. In a two-beam extension its contribution would have its own polarization, detuning, and branching channels; replacing γ by $\gamma + \gamma_r$ is sufficient only if it has the same reduced spin-flip structure. The next model instead uses one beam for both control and measurement. Omitting `--eta` in v5 sets the dissipative map to the identity and recovers the v4 closed-system path.

3 One beam without manifold loss: v6

The same probe now generates the tensor shift, scatters, and supplies the Faraday readout. Sweeping its strength must change the noise as well as the relaxation:

$$\gamma = \frac{\beta}{\eta}, \quad \gamma \sigma^2 \delta t = \frac{1}{\text{OD}_{\text{eff}} N_a}, \quad \sigma^2(\beta) = \frac{\eta}{\text{OD}_{\text{eff}} N_a \delta t \beta}. \quad (8)$$

Here OD_{eff} absorbs the conversion between spin-flip relaxation and Faraday measurement strength; it is not automatically the resonant optical depth. The dataset `data_mle_over_time_onebeam_1` uses $\eta = 30$, $\text{OD}_{\text{eff}} = 10$, and $N_a = 2 \times 10^5$.

Increasing β strengthens nonlinear dynamics and reduces shot noise, but also shortens the useful signal duration. At $\beta = 0$ there is no probe and σ^2 diverges, so `v6/v7` exclude zero from the sweep. To isolate the tensor term, `--tensor-scale 0` sets $\beta_H = 0$ while retaining the rates and noise of a nonzero β . This is a numerical control at fixed measurement strength, not a zero-intensity experiment.

The illustrative envelope $\partial s / \partial w \sim t e^{-\gamma t}$ gives $F \sim \sigma^{-2} \int_0^T t^2 e^{-2\gamma t} dt$. Information initially grows with probe strength but saturates as sensitivity decays, suggesting a scale $\beta \sim \eta / T$. This is a heuristic rather than an exact optimum for the three-parameter random-control problem. Direction-dependent sensitivity and the tensor dynamics must be computed.

4 One beam with Raman loss and atomic coefficients: v7

4.1 Rates and signed tensor shift

The current model adds Raman transfer from the chosen ground hyperfine manifold to the other manifold. The supporting calculation `cs_light_shift_scattering_ratios.py` constructs cesium dipole matrices from angular-momentum coefficients and the atomic data conventions of Ref. [2]. For a linearly polarized (π) probe it adiabatically eliminates one excited fine-structure manifold, checks dipole closure and scattering normalization, and extracts the tensor shift, within-manifold relaxation, loss operator, and vector-to-scalar polarizability ratio.

The `v7` script uses rounded presets from that calculation at detuning -20 GHz from the selected ground-manifold-to-excited-centroid transition. They are effective far-detuned coefficients, not a full detuning-dependent pumping model. Let γ_s be the scattering-rate scale normalized to the initial $+x$ coherent state, and $q_v = b/a$ the vector-to-scalar polarizability ratio per unit magnetic quantum number. Then

$$\gamma_s = \frac{\beta}{|\eta_s|}, \quad \gamma_{\text{sf}} = r_{\text{within}} \gamma_s, \quad (9)$$

$$\beta_H = s_T \text{sgn}(\eta_s) \beta, \quad K_L = s_L \gamma_s (a_L \mathcal{K} + c_L J_z^2), \quad (10)$$

$$\sigma^2 = \frac{1}{q_v^2 \text{OD}_{\text{res}} N_a \delta t \gamma_s}, \quad \text{OD}_{\text{res}} = \frac{N_a \sigma_{\text{res}}}{A_{\text{beam}}}. \quad (11)$$

The CLI sweep variable β is positive; η_s supplies the tensor-shift sign. The flags `--tensor-scale` and `--loss-scale` set s_T and s_L , both one by default. The resonant cross section must use the same normalization as the Faraday-noise convention in Eq. (11).

Preset	η_s	r_{within}	a_L	c_L	q_v
Cs D2, $F = 3$	7.37	0.0055	0.204	0.0107	0.127
Cs D2, $F = 4$	-9.55	0.0052	0.130	0.0104	-0.123
Cs D1, $F = 3$	-100.3	0.0110	0.412	0.0264	-0.243
Cs D1, $F = 4$	122.0	0.0115	0.254	0.0170	0.257

Table 2: Presets stored in the `v7` script. All four cluster runs analyzed here use Cs D2, $F = 3$. The other presets are available, but were not simulated in these datasets.

The coefficient r_{within} multiplies the $J_{\pm}$ dissipator. It is a normalized transverse-relaxation coefficient, not the probability of a within-manifold event per scattered photon. Likewise γ_s is a fixed reference scale; instantaneous scattering of an evolving state need not equal its initial-state value.

4.2 Trace-decreasing master equation and measured signal

Using Eq. (5) at rate γ_{sf} , the reduced state obeys

$$\dot{\rho} = -i[H, \rho] + \mathcal{D}_{\text{sf}}[\rho] - \frac{1}{2}\{K_L, \rho\}. \quad (12)$$

Here $K_L = \sum_q (L_q^L)^\dagger L_q^L$ for transfer operators L_q^L into the other manifold. Their recycling terms lie outside the modeled space and are omitted. Thus

$$p = \text{Tr } \rho, \quad \dot{p} = -\text{Tr}[K_L \rho], \quad \text{Tr}[O\rho] = p \text{Tr}[O\rho_{\text{cond}}], \quad \rho_{\text{cond}} = \rho/p. \quad (13)$$

The unnormalized expectation is the signal per initial atom if transferred atoms are dark. It must not be renormalized to the surviving population before adding the noise in Eq. (2).

Loss is state dependent because $c_L \neq 0$. Even without spin flips, conditioning on survival changes the state through non-Hermitian attenuation; it need not follow the lossless unitary trajectory. Only uniform loss $K_L = \kappa \mathbb{I}$ factors into $e^{-\kappa t}$ times otherwise unchanged dynamics. In particular, $e^{-\langle K_L \rangle_0 t}$, used as a survival estimate in the notebook's initial rate table, is not an exact survival curve for the driven model.

For the initial $+x$ coherent state, $\langle J_z^2 \rangle_0 = J/2$. The initial *dissipative* relaxation rate of $\langle J_x \rangle$ is

$$R_0 = \gamma_{\text{sf}} + \langle K_L \rangle_0 = \gamma_s [r_{\text{within}} + s_L(a_L + c_L J/2)]. \quad (14)$$

This is saved as `gamma_array.npy`; it differs from `gamma_sf_array.npy`. For Cs D2, $F = 3$, with $s_L = 1$,

$$\langle K_L \rangle_0 = 0.22005\gamma_s, \quad R_0 = 0.22555\gamma_s = 0.0306038\beta, \quad \beta/R_0 = 32.68. \quad (15)$$

Loss contributes 97.56% of this initial relaxation. This percentage is not the branching fraction of all scattering events, and R_0^{-1} is a reference time scale, not a universal exponential decay time.

4.3 Relation to the previous one-beam run

Identifying the v6 rate with R_0 gives the noise-matching convention

$$\text{OD}_{\text{eff}} = \frac{q_v^2}{R_0/\gamma_s} \text{OD}_{\text{res}} = 0.07151 \text{OD}_{\text{res}} \quad (\text{Cs D2}, F = 3). \quad (16)$$

Thus OD 140 approximately reproduces the earlier effective OD 10. The actual comparison is close but not exact: v6 uses $\beta/\gamma = 30$, v7 has $\beta/R_0 = 32.68$, and the v7 noise standard deviation is 4.3% larger. These are nearby operating points with different dissipators, not an exact matching of every effective quantity.

As an algebraic check, $s_L = 0$ removes loss. For $s_T = 1$, positive η_s , and overrides satisfying $r_{\text{within}}/\eta_s = 1/\eta$ and $q_v^2 \text{OD}_{\text{res}}/\eta_s = \text{OD}_{\text{eff}}/\eta$, the v7 equations reduce to v6. Setting only $s_T = 0$ preserves both scattering and noise.

5 Numerical propagation and parameter fitting

The adjoint irreversible generator is

$$\mathcal{D}^\dagger[O] = -\gamma_{\text{sf}} \sum_{k=x,y} [J_k, [J_k, O]] - \frac{1}{2} \{K_L, O\}. \quad (17)$$

For row-major vectorization, $\text{vec}_C(XOY) = (X \otimes Y^\top) \text{vec}_C(O)$. The script exponentiates this superoperator to obtain $\mathcal{A} = \exp(\delta t \mathcal{D}^\dagger)$, a 49×49 matrix at $J = 3$. It is constant within a realization and independent of $\mathbf{w}$.

With $U_w = e^{-i\delta t \mathbf{w} \cdot \mathbf{J}}$ and $U_c[n] = e^{-i\delta t H_c[n]}$, the implemented update is

$$O_{n+1} = U_w^\dagger U_c[n]^\dagger \mathcal{A}[O_n] U_c[n] U_w. \quad (18)$$

The dissipative substep is exact; separating dissipation from the unitaries, and field from control, are first-order splittings. Without spin flips, $\mathcal{A}[O] = e^{-\delta t K_L/2} O e^{-\delta t K_L/2}$. With both channels present, $\mathcal{A}[\not K]$ is generally not $e^{-\delta t K_L}$, since the spin-flip and loss maps need not commute.

Time-ordering convention. Successive adjoint maps act on the observable in increasing array-index order. If $\mathcal{E}_n$ is the dual state map, after two steps

$$\text{Tr}[\mathcal{E}_1^\dagger \mathcal{E}_0^\dagger(O_0)\rho_0] = \text{Tr}[O_0 \mathcal{E}_0 \mathcal{E}_1(\rho_0)]. \quad (19)$$

This reverses the control order relative to forward Schrödinger propagation with that indexed sequence. Data generation, residuals, and Fisher derivatives all share this convention. The results describe that implemented protocol; identifying its entire record with a chronologically driven experiment requires a separate ordering check. Single-step duality alone does not establish equality of the complete time-ordered records.

Fréchet derivatives of U_w generate $\partial O_n / \partial w_a$ by the product rule; $\mathcal{A}$ acts on both the observable and its derivatives. For known control realization i the classical Fisher matrix is

$$F_{i,ab}(T) = \sigma^{-2} \sum_{n \leq T} \frac{\partial s_{i,n}}{\partial w_a} \frac{\partial s_{i,n}}{\partial w_b}. \quad (20)$$

Atomic coefficients, optical depth, atom number, and noise are treated as known.

At each cutoff the estimator minimizes $C = \frac{1}{2} \sum_n (M_n - s_n)^2$ using SciPy's bounded trust-region least-squares method with $-1 \leq w_a \leq 1$. Starting from $\mathbf{w} = 0$, it fits successively longer prefixes (250 steps, then doubling, ending at the cutoff). Cutoffs do not share fitted estimates. An excessive final cost triggers up to two random restarts, keeping the lowest-cost fit. Controls, noise, and restart points are drawn before joblib dispatch. Each realization reuses its record across cutoffs.

6 Interpretation of variances, filtering, and Fisher bounds

For a fixed true field and known control, the local unbiased-estimator bound is $\text{Cov}(\hat{\mathbf{w}} \mid i) \succeq F_i^{-1}$. Two averages must be distinguished:

$$B_{\text{cond}} = \mathbb{E}_i[F_i^{-1}], \quad B_{\text{avg}} = (\mathbb{E}_i[F_i])^{-1}, \quad B_{\text{cond}} \succeq B_{\text{avg}}. \quad (21)$$

The inequality assumes positive-definite information matrices. The conditional bound is the benchmark for locally unbiased estimates made separately for each control record. Inverse average

information instead pertains to an averaged-information experiment, such as joint estimation from many records with the appropriate record-count factor. It is not the bound for an arbitrary fixed control sequence.

For conditionally unbiased estimates, total covariance equals the average conditional covariance. Control-dependent bias, finite samples, parameter bounds, and selection complicate the comparison. Agreement with B_{cond} is consequently an empirical diagnostic, not proof of unbiasedness or global identifiability.

The script retains all 300 fits. The notebook subsequently rejects those above

$$C_{\text{band}} = \frac{\sigma^2}{2}(n + 5\sqrt{2n}), \quad \mathbb{E}[C(\mathbf{w}_{\text{true}})] = \frac{n\sigma^2}{2}, \quad (22)$$

where n counts measurements at the cutoff. This noise-based threshold is referenced to the cost at the truth; passing it does not guarantee the correct minimum. Surviving fits give sample variances with denominator $N_{\text{kept}} - 1$. The notebook computes statistics separately for each true-field draw, then averages the summaries across 20 tasks. It does not pool estimates of different true fields into a single covariance.

The plotted quantity is filtered total variance $V(T) = \sum_a \text{Var}(\hat{w}_a)$, excluding squared bias. Both Fisher summaries retain all realizations, so the empirical variance and bounds refer to different selected populations when fits are rejected. Mean rejection fractions over the beta/time grid are 3.0%, 3.8%, 4.5%, and 5.6% for OD 140 on/off and OD 10 on/off, respectively; the largest task-averaged fraction at a single setting is 13.2%. Dashed curves show $\text{Tr } B_{\text{cond}}$ and dotted curves $\text{Tr } B_{\text{avg}}$. The time-to-target metric is filtered variance, not total mean-squared error.

7 Cluster configuration and v7 results

7.1 Data provenance and paired controls

The cluster simulations use `multiparameter_trotter_v7_loss_least_squares_over_time.py`; analysis is in `multiparameter_plots_v7.ipynb`. Each dataset has 20 array tasks with seeds 0–19. Each task draws one field with independent $w_a \sim \mathcal{N}(0, 0.1^2)$ and runs 300 realizations for every $\beta \in \{0.5, 1, 2, 5, 10, 20\}$. The step is $\delta t = 10^{-3}$, duration $T = 14$, and 40 integer cutoffs span steps 10–14,000. Measurement indices are generated by the script’s integer `linspace(0,14000,14000)`, including step zero; the notebook uses the same indices to count data points.

Dataset suffix	OD _{res}	s_T	Purpose
<code>loss_1</code>	140	1	Reference optical depth
<code>loss_1_notensor</code>	140	0	Tensor-off control
<code>loss_1_od10</code>	10	1	Lower optical depth
<code>loss_1_od10_notensor</code>	10	0	Tensor-off control

Table 3: Directory prefix: `data_mle_over_time_`. All runs use `--atom Cs-D2-F3`, $N_a = 200,000$, and $s_L = 1$. Rates and shot noise are identical within each tensor-on/off pair.

Matched seeds and draw ordering pair the true fields and control phases across tensor-on/off runs, as confirmed by the saved arrays. Noise is paired by the generation procedure. Reducing OD from 140 to 10 changes only the noise scale at fixed β , increasing σ by $\sqrt{14}$. At $\beta = 1$, $\gamma_s = 0.13569$, $\gamma_{\text{sf}} = 0.0007463$, $\langle K_L \rangle_0 = 0.02986$, and $\sigma = 0.12775$ (OD 140) or 0.4780 (OD 10). The lower optical

depth represents a less favorable readout setting; a quantitative beam-size interpretation requires a specified cross-section and mode-area convention.

7.2 Variance relative to the conditional benchmark

Figure 1 compares the four datasets. At $T = 14$ in the OD 140 tensor-on run, $V/\text{Tr } B_{\text{cond}} = 0.99\text{--}1.03$ across all six probe strengths. This is close agreement for the filtered estimator at that operating point. It does not establish efficiency for every time or configuration. For example, in the OD 10 tensor-on run this ratio is about 17 at $\beta = 0.5$ and 6.6 at $\beta = 20$, while intermediate strengths remain near one.

Tensor-off estimates can also remain far above their conditional bounds: the final ratio at $\beta = 20$ is about 17 for OD 140 and 92 for OD 10. A large gap can reflect nonlinear estimation errors, weak identifiability, bias, or optimizer behavior missed by the cost test. These data do not uniquely diagnose the cause. Conditional Fisher comparisons help distinguish local information gain from the additional estimator-performance gap.

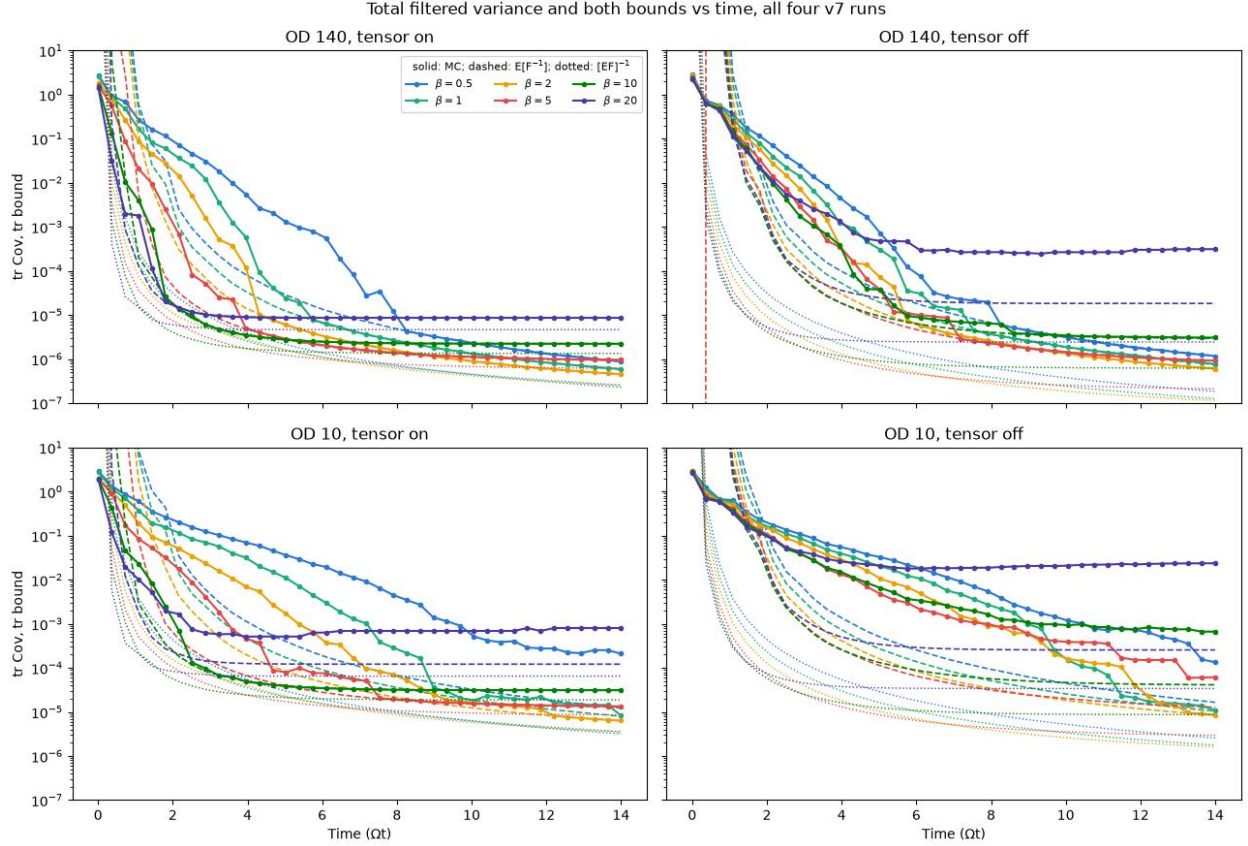


Figure 1: Existing v7 analysis figure: total filtered variance (solid), $\text{Tr } E_i[F_i^{-1}]$ (dashed), and $\text{Tr}(E_i[F_i])^{-1}$ (dotted). Tensor on/off occupy the left/right columns; OD 140/10 occupy the top/bottom rows. Curves average summaries over 20 true-field draws.

7.3 Time to a target filtered variance

At each fixed β , the notebook finds the last cutoff above a target and interpolates the next crossing linearly in time and logarithmically in variance. It then chooses the fastest result over the six strengths. “Stays below” means below at subsequent sampled cutoffs up to $T = 14$; no claim is made about unsampled times or later evolution. The chosen β is constant throughout the record, not changed during acquisition.

OD	Tensor	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}
140	on	0.52 (20)	1.16 (20)	1.47 (20)	2.46 (10)	9.55 (2)
140	off	2.13 (10)	3.28 (10)	4.27 (10)	5.69 (2)	11.21 (2)
10	on	1.09 (20)	2.06 (10)	2.85 (10)	11.99 (2)	–
10	off	4.37 (5)	7.77 (2)	10.67 (1)	13.24 (2)	–

Table 4: Best interpolated time Ωt to each target V , with the fixed β in parentheses. A dash means not reached during the run. Values are notebook point estimates without uncertainty intervals.

The tensor term shortens the best time for every target reached by both members of a pair. At OD 140 the speedup ranges from about 4.1 for 10^{-2} to 1.17 for 10^{-6} . At OD 10 it is 3.7–4.0 for 10^{-2} through 10^{-4} , but only 1.10 for 10^{-5} . The largest improvement is in reaching moderate precision sooner, rather than a uniform speedup at every precision.

7.4 Tensor contribution at fixed probe strength

Figure 2 compares equal β , with the same scattering and noise. The early-time gains can be substantial, but the tensor term does not win at every fixed strength and time. At $T = 14$, the conditional-bound ratios (on/off) for increasing β are

$$(0.741, 0.770, 0.749, 1.020, 0.756, 0.475). \quad (23)$$

The $\beta = 5$ setting is a small exception: the tensor-on bound is about 2% higher, and its OD 140 filtered variance is about 5% higher. At OD 10, $\beta = 0.5$ also has larger final tensor-on variance despite a lower conditional bound. These exceptions coexist with the optimized time-to-target gains.

Inverse-average-information ratios at $T = 14$ instead range from 1.24 to 2.90, favoring tensor-off dynamics. Averaging before inversion weights poorly informative control realizations differently. For these final-time data the Jensen gap relative to B_{avg} is larger in the tensor-off case, which explains the reversed ranking of the two bounds. Very large Monte Carlo gains additionally include settings where tensor-off fits do not reach their own conditional benchmark.

7.5 Comparison with spin-flip-only relaxation

Figure 3 compares v7 OD 140 tensor-on with v6 `onebeam.1`. At $T = 14$, filtered-variance ratios v7/v6 are approximately 0.84, 0.44, and 0.075 at $\beta = 0.5, 1, 2$; at $\beta \geq 5$ they are of order 10^{-3} . The loss-dominated reduced model therefore performs much better than the spin-flip-only model at these approximately matched operating points.

A single initial relaxation rate is insufficient to predict estimation performance. This comparison does not mean that adding loss to a fixed spin-flip model improves precision: v7 reduces the spin-flip coefficient while introducing loss. It also does not imply unitary dynamics of survivors, because attenuation is state dependent. The figure’s right-hand panel compares B_{avg} , as in the notebook, rather than conditional efficiency.

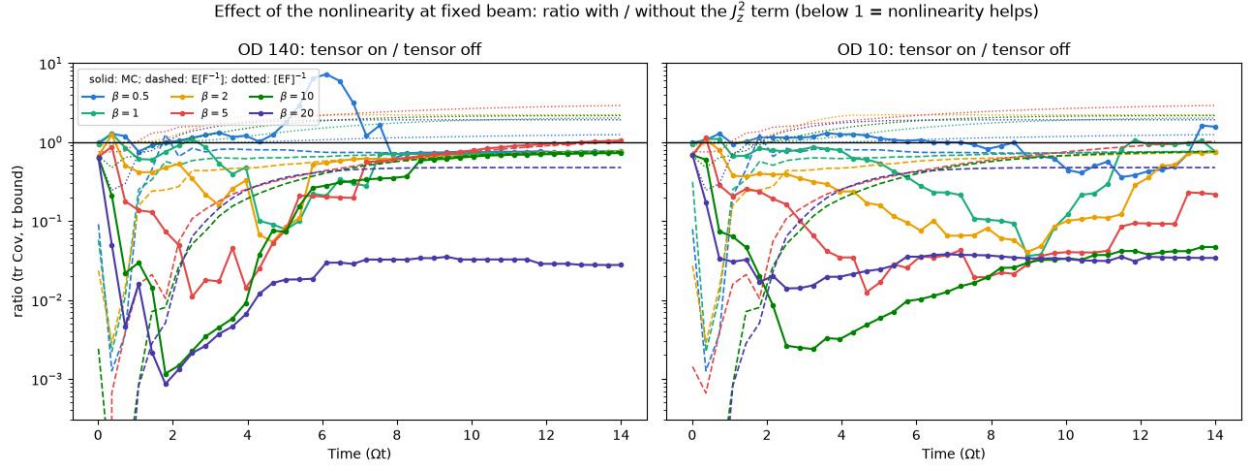


Figure 2: Tensor-on/tensor-off ratios at fixed nonzero probe strength. Solid: filtered variance; dashed: conditional Fisher benchmark; dotted: inverse average information. Values below one favor the tensor term.

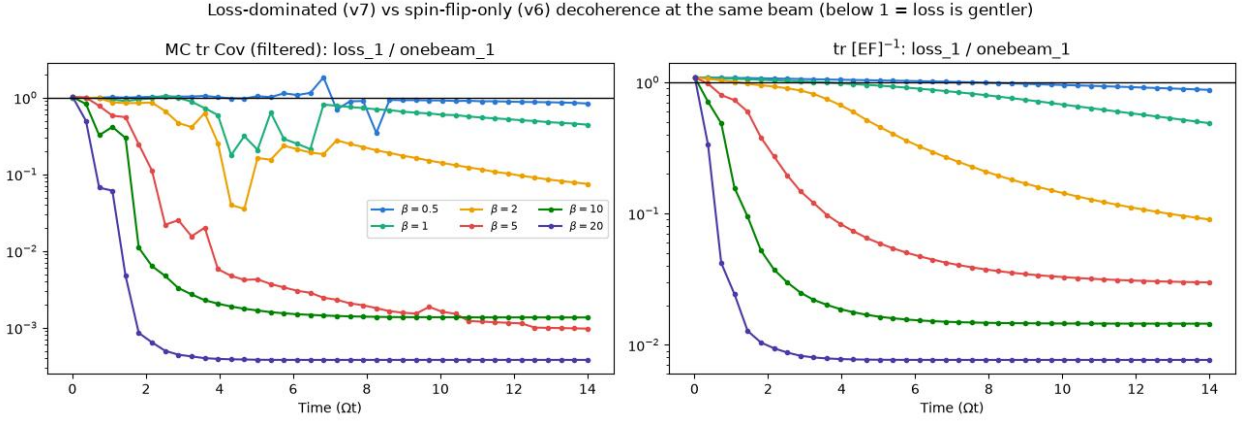


Figure 3: Existing comparison of v7 `loss_1` with v6 `onebeam_1`: filtered variance ratio (left), inverse-average-information ratio (right). The figure's “same beam” wording denotes nearby operating points; their noise and initial relaxation rates differ as quantified in the text.

At OD 10 the Hamiltonian and irreversible dynamics are unchanged. For matched controls both Fisher bounds scale exactly by 14, since only σ^2 changes; the tensor-on/off Fisher ratios are consequently independent of OD. Fitted variance need not scale exactly, because noise changes estimator behavior and filtering. The lower-depth runs preserve the optimized advantage in Table 4, but do not reach $V = 10^{-6}$ within $T = 14$.

8 Scope of the current model and next checks

The current configuration is a single-beam, single-manifold mean-signal model with atom-derived coefficients, spin flips, and state-dependent Raman loss. The conclusions above concern its paired v7 datasets and implemented control convention. The following extensions would strengthen the physical and statistical interpretation.

1. **Both ground manifolds.** Transferred atoms are treated as dark and never return. A fuller cesium model would evolve $F = 3 \oplus 4$ (dimension 16), include inter-manifold pumping and both optical responses, and use a 256×256 operator space.
2. **Measurement fluctuations.** The likelihood adds independent photon shot noise to a deterministic ensemble mean. Atomic projection noise, stochastic loss-number fluctuations, conditional measurement back-action, and induced interatomic correlations are not separately simulated. Faraday coupling and optical depth also require calibration for the detection geometry.
3. **Ordering and numerical convergence.** Compare the convention in Section 5 with forward state propagation before interpreting records as a chronological experiment. Quantify splitting error through a step-size convergence study.
4. **Estimator diagnostics.** Compare unfiltered variance, bias, mean-squared error, rejection fractions, and conditional information. Add uncertainty estimates and targeted checks of restarts and identifiability before assigning a unique cause to the tensor-off excess variance.
5. **Broader optimization.** The results cover six constant probe strengths, one atomic preset, two optical depths, and $T \leq 14$. Other presets and time-dependent probe schedules have not been tested by these four datasets.

A Repository map and reproducibility

The root-level v5/v6 decoherence scripts implement the earlier stages; the v7 script is identified in Section 7. Matching shell launchers specify `--od-res 140` or `10` and the `--tensor-scale 0` controls. Each `task_N/parameters.txt` is the primary record of its actual run configuration.

In v7, `gamma_s_array.npy`, `gamma_sf_array.npy`, `loss_rate_scs_array.npy`, and `sn_sd_array.npy` record the separate scales. `w_est.npy` and `cost_est.npy` retain all fitted records; `phi_kept.npy` stores compressed phases. `fisher.npy` stores average information, whereas `fisher_inv_realavg.npy` stores averaged individual inverses. Raw fit arrays have leading axes trial, beta, cutoff, realization; Fisher summaries have already averaged over realizations.

The v7 notebook filters and averages directly from the task folders. The three PDFs under `latex/figures/` are existing analysis figures. Numerical summaries in this note were checked against the saved arrays using the notebook’s aggregation and interpolation rules, without new cluster simulations. Compile from `latex/` with `latexmk -pdf open_system_model.tex`.

References

- [1] I. H. Deutsch and P. S. Jessen, “Quantum control and measurement of atomic spins in polarization spectroscopy,” *Optics Communications* **283**, 681–694 (2010). doi:10.1016/j.optcom.2009.10.059; arXiv:0909.4506.
- [2] D. A. Steck, *Cesium D Line Data*, revision 2.3.4 (8 August 2025), <https://steck.us/alkalidata/>.